GG–C–139
21 JULY 1949

FEDERAL SPECIFICATION

CATHETERS, URETHRAL, WOVEN (BRAIDED)

This specification was approved by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government

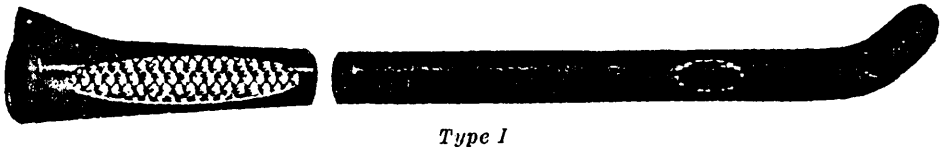
1. CLASSIFICATION

1.1 Types. — Woven (braided) urethral catheters covered by this specification shall be of the following types, as specified:

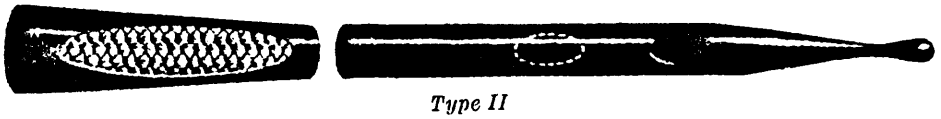
Type I.—Mercier, Coude tip.
Type II.—Straight, olive tip.
Type III.—Straight, Philips, with threaded tip.
Type IV.—Double eye, modified natural curve, olive tip.

1.2 Sizes.—The urethral catheters shall be of the following French scale sizes for the respective types, as specified:

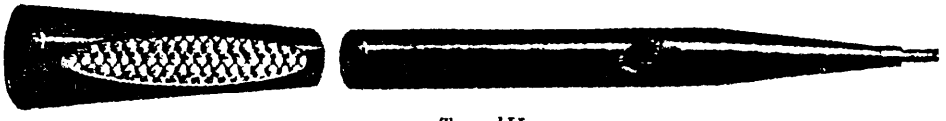
For type I: 4Fr to 30Fr, inclusive, even sizes.
For type II: 4Fr to 30Fr, inclusive, even sizes.
For type III: 8Fr to 30Fr, inclusive, even sizes.
For type IV: 8Fr to 30Fr, inclusive, even sizes.



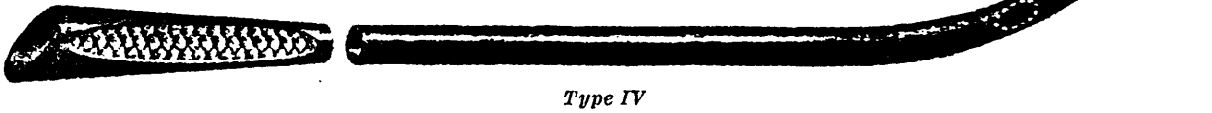
Type I



Type II



Type III



Type IV

Figure 1.—*Woven urethral catheters*

2. APPLICABLE SPECIFICATIONS

2.1—There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

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3. REQUIREMENTS

3.1 Materials.

3.1.1 *Yarn.*—Unless otherwise specified, the yarn shall be good quality cotton, silk, linen, or nylon, or other synthetic yarn. The nylon or other synthetic yarn shall have a melting point of not less than 135° C. when tested as described in 4.3.6.

3.1.2 *Coating.*—The coating shall be a flex-

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ible varnish or suitable synthetic or plastic compound.

3.2 General description.—The catheters shall be composed of a tightly and smoothly braided base, uniformly covered with a flexible coating. The finished catheter shall have the proper flexibility, pliability, and body required for the intended use. The proximal ends shall be flared to fit the connection or tapered tip of an irrigating syringe. The external surface shall be smoothly textured and finely finished. The catheters shall be free from toxic material or any ingredient which may have a deleterious effect upon body tissues.

3.2.1 *Base.*—The base shall be a suitable tube, tightly and smoothly braided of yarn of appropriate size and type for the size of the catheter. The interior shall be smooth, well rounded, and nonabsorbent. Each yarn in the tube shall be of one continuous length.

3.2.2 *Eyes.*—The eyes shall be smoothly and symmetrically braided into the structure of the tube. The area of each eye shall be not less than the cross-sectional area of the catheter.

3.2.3 *Coating.*—The braided base shall be thoroughly and uniformly impregnated with a coating as specified in 3.1.2, designed to give maximum life in repeated use under actual working conditions. The coating shall thoroughly cover the base, and the surface shall be without irregularities or imperfections of any kind.

3.2.3.1 *Chemical properties.*—The chemical properties of the coating shall be such that when tested in accordance with 4.3.2, 4.3.3, 4.3.4, and 4.3.5, the catheters will not stick together or show signs of brittleness, blistering, swelling, peeling, cracking, or any other damage.

3.2.4 *Diameters.*

3.2.4.1 *Outside diameter.*—The outside diameter of the main shaft of the catheter beyond the flare at the proximal end shall pass through the specific scale opening of the size designated but not through the next smaller size opening of the French scale when tested as described in 4.3.1. (The diameter, when expressed in millimeters according to the French scale, is numerically equal to one-third the scale size.) The outside diameter of the catheter shall be determined by that portion of the main shaft between the beginning of the flare at the proximal end and the beginning of the taper at the distal end.

3.2.4.2 *Inside diameter.*—The inside diameter of the catheters shall be as shown in table I for the respective sizes.

TABLE I.—*Inside diameter*

French scale sizes	Inside diameter, minimum
	Millimeters
4 Fr	0.41
6 Fr	0.84
8 Fr	1.18
10 Fr	1.50
12 Fr	1.73
14 Fr	2.29
16 Fr	2.44
18 Fr	3.02
20 Fr	3.58
22 Fr	4.12
24 Fr	4.75
26 Fr	5.21
28 Fr	5.71
30 Fr	6.22

3.2.5 *Finish.*—The exterior surface of the catheters shall be properly worked down to give a dry, smooth, highly polished, flexible finish. The surface shall be "slick" when wet.

3.2.6 *Marking.*—Each catheter shall be plainly and indelibly marked upon the proximal end with the name or registered trademark of the manufacturer and the French size of the catheter.

3.3 Type I.—Type I catheters shall be of the Mercier Coude tip design similar to that shown in figure 1. The proximal end may be furnished with an angle as shown in figure 1 for type IV. The distal eye shall be close to the distal tip. The distal tip shall be well-rounded, and the two eyes shall be spaced so as to allow a smooth curve between them. The curve shall be a 35° ± 10° angle. The catheters shall be 13½ ± ½ inches in length.

3.4 Type II.—Type II catheters shall be straight with an olive tip, similar to the design shown in figure 1. The catheters shall be uniform in diameter from the ending of the flare to just beyond the distal eye. The diameter of the portion beyond the distal eye, adjacent to the olive tip, shall taper not more than 60 percent of the diameter of the main shaft in sizes 10Fr to 12Fr, inclusive, 55 percent in sizes

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14Fr to 22Fr, inclusive, and not less than 35 percent in all sizes. The catheters shall be 13½ ± ½ inches in length.

3.5 **Type III.**—Type III catheters shall be similar to the Philips design shown in figure 1. The catheters shall have an American National Standard screw-thread tip, 0.073 inch in diameter, 56 external threads (No. 1 fit) per inch, of a length sufficient to screw an internal fitting of like thread eight full turns. The tip shall be made of monel metal or of nickel silver. The catheters shall be 13½ ± ½ inches in length.

3.6 **Type IV.**—Type IV catheters shall be similar to the design shown in figure 1. The catheters shall have a modified natural curve, double eye, and olive tip. The catheters shall be uniform in diameter from the end of the flare to just beyond the distal eye. The diameter of the portion beyond the distal eye adjacent to the olive tip shall taper not more than 60 percent of the diameter of the main shaft in sizes 10Fr to 12Fr, inclusive, 55 percent in sizes 14Fr to 22Fr, inclusive, and not less than 35 percent in all sizes. The catheters shall be 13½ ± ½ inches in length.

3.7 **Workmanship.** — The workmanship shall be first class throughout. The finished catheters shall be symmetrical in shape and shall be free from any irregularities or imperfections which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—Unless otherwise specified, at least four catheters of each type and size shall be taken at random from each lot of 200 or fraction thereof delivered.

4.2 **Inspection.** — Samples shall be inspected for material, workmanship, design, markings, dimensions, and finish.

4.3 **Tests.**

4.3.1 *Diameters.*

4.3.1.1 *Outside diameter.*—The outside diameter of all the catheters taken for tests shall be tested for size with a standard French scale.

4.3.1.2 *Inside diameter.*—The inside diameter of all samples shall be tested by passing a mandrel of appropriate diameter through the catheter to the eye nearest the proximal end. The diameter of the mandrel shall be not less than that shown in table I for minimum inside diameter.

4.3.2 *Boiling test.*—One catheter of each type and size shall be boiled for ten separate periods of 3 to 5 minutes each. Between periods the catheters shall be removed from the water, dried for 1 hour at 200° F., and then cooled to room temperature (68° to 72° F.).

4.3.3 *Autoclaving test.*—One catheter of each type and size shall be autoclaved for ten 10-minute periods at 250° F. Between periods the catheters shall be dried for 1 hour at 200° F. and then cooled to room temperature (68° to 72° F.).

4.3.4 *Cleaning test.*—One catheter of each type and size shall be cleaned for 5 minutes with 70 percent alcohol.

4.3.5 *Dry heat test.*—One catheter of each type and size shall be subjected to oven heat for 15 minutes at 250° F., and then allowed to return to room temperature (68° to 72° F.).

4.3.6 *Melting point.*—A piece of synthetic yarn approximately ⅛ inch long, taken from the base, shall be placed in a capillary tube closed at the lower end. The piece of yarn shall be attached to a partial immersion thermometer conforming to A. S. T. M. designation E1 (2C–39) (−5° to +300° C.) (requirements for A. S. T. M. standard thermometers are contained in Fed. Spec. VV–L–791 under method No. 950.1) and immersed to the proper depth in a bath at a temperature of 135° C. for 5 minutes.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.**—Unless otherwise specified, each catheter shall be packaged in accordance with accepted commercial practice.

5.2 **Packing.**—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate and to afford maximum protection from normal hazards of transportation.

5.3 **Marking.**

5.3.1 *Packages.*—Unless otherwise specified, packages shall be marked with name of the commodity, the type, size, quantity contained therein, and the names of the contractor and manufacturer.

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5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name, type, and size of the commodity as defined by the contract or order under which the shipment is made, the quantity of each type contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein and should specify the type and size required. (See 1.1, 1.2, 3.1.1, 4.1, 5.1, 5.3, and 7.2.5.)

6.2 This specification covers the following items listed in the Army-Navy Catalog of Medical Materiel:

3–180–100, Catheter, Urethral, Woven, Mercier 8Fr.
3–180–130, Catheter, Urethral, Woven, Mercier 18Fr.
3–180–140, Catheter, Urethral, Woven, Mercier 16Fr.
3–180–150, Catheter, Urethral, Woven, Mercier 18Fr.
3–180–160, Catheter, Urethral, Woven, Mercier 20Fr.
3–180–180, Catheter, Urethral, Woven, Mercier 24Fr.
3–180–740, Catheter, Urethral, Woven, Straight 16Fr.
3–180–750, Catheter, Urethral, Woven, Straight 18Fr.
3–181–300, Catheter, Urethral, Woven, Straight, Philips 12Fr.
3–181–310, Catheter, Urethral, Woven, Straight, Philips 14Fr.
3–181–700, Catheter, Urethral, Woven, Double Eye 16Fr.
3–181–710, Catheter, Urethral, Woven, Double Eye 18Fr.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine conformance to the requirements of this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army and Navy.

7.2.1 *Applicable specifications and other publication.*—

Federal Specifications:

NN–B–601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.
NN–B–621—Boxes; Wood, Nailed and Lock-corner.
NN–B–631—Boxes; Wood, Wirebound, for Domestic Shipment.

National Military Establishment Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

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JAN-B-107—Boxes, Wood, Wirebound (Overseas Type).
JAN-P-120—Packaging and Packing for Overseas Shipment—Cartons, Folding, Paperboard.
JAN-P-125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
JAN-P-133—Packaging and Packing for Overseas Shipment—Boxes; Set-Up, Paperboard.
JAN-P-140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

U. S. Army Specification:

100-2—Standard Specification for Marking Shipments by Contractors.[1]

Navy Department Specification:

General Specifications for Inspection of Material.[2]

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.[2]

(Copies of specifications required for Army purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.

(Copies of Federal and National Military Establishment specifications required for Navy purchases, Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Virginia. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.)

7.2.2 *Packaging, domestic and overseas.*

7.2.2.1 *Unit packages.*—Each catheter shall be packaged (not coiled) in a glassine or waxed paper envelope and sealed. As an alternative, each catheter may be wrapped (not coiled) in waxed or glassine paper and sealed.

7.2.2.2 *Intermediate packages.*—Six units shall be further packaged in a snug-fitting, folding, paperboard carton or set-up paperboard box conforming to the requirements of NME Specifications JAN-P-120 and JAN-P-133, respectively. The carton shall be adequately secured to prevent accidental opening.

7.2.3 *Packing.*

7.2.3.1 *Domestic.*—Sixty intermediate packages, 360 units, shall be packed in a shipping container conforming to the requirements of Federal Specification NN-B-601, NN-B-621, or NN-B-631.

7.2.3.2 *Overseas.*—Sixty intermediate packages, 360 units, shall be packed for type 2 load in a shipping container conforming to the requirements of NME Specification JAN-P-105, JAN-P-106, or JAN-B-107. Closure and strapping shall comply with the appendix of the applicable specification. Each box shall be lined with a sealed waterproof bag or its equivalent, made from material conforming to the requirements of NME Specification JAN-P-125, for case liners; seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of NME Specification JAN-P-140.

7.2.3.3 *Packing variation permitted.*—If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in a shipping container of suitable size and design and so constructed as to insure safe delivery to specified destination.

7.2.4 *Marking.*

7.2.4.1 *Interior packages.*—Each unit package and intermediate package shall be marked with the stock number, nomenclature, unit, quantity, and the name of the contractor.

7.2.4.2 *Shipping containers.*

7.2.4.2.1 *Army.*—Each shipping container shall be marked in accordance with the requirements of U. S. Army Specification 100-2.

7.2.4.2.2 *Navy.*—Each shipping container shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook.

7.2.5 *Ordering data.*—Requests, requisitions,

[1] Applicable to Army purchases only.
[2] Applicable to Navy purchases only.

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schedules, and contracts or orders should specify the following:

(*a*) Domestic or overseas packing. (See 7.2.3.)

NAVY: MS

(*b*) Domestic or overseas marking of shipping containers. (See 7.2.4.2.)

(*c*) Army or Navy marking of shipping containers. (See 7.2.4.2.)

This specification is part of Section IV, Part 5, of the Federal Standard Stock Catalog

For sale by the Superintendent of Documents, Washington 25, D. C. Price 10 cents.

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